

Study Notes: “Investing in Misallocation” (Kilic & Tuzel, JFE 2026)

Notes (models + interpretation of all tables/figures)

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1 One coherent thread: the paper’s question, mechanism, and why it matters

1.1 The empirical puzzle the paper starts from

The paper begins with a misallocation-style observation: in the cross-section, some firms have *high investment rates* but *low marginal products of capital* (low MPK). In a static efficiency logic (or in a standard neoclassical model without special future opportunities), this looks like “too much capital” allocated to low-return firms.

The key empirical regularity is that the *high- I/K , low-MPK* group is not rare. In the portfolio sorts used throughout the paper, the $(I/K_2, MPK_1)$ portfolio is about one-fifth of all firms, and it is *even more common among young firms*. Moreover, these firms are much more likely to experience extreme growth episodes (“jumps”) than the typical firm.

1.2 Core interpretation

The paper’s interpretation is that the high- I/K , low-MPK pattern can be *efficient* and forward-looking: firms may invest not only for current production, but also to increase the probability of future large productivity improvements (innovation/product development/scale expansion).

This is the organizing logic:

High investment with low current MPK can be optimal if investment increases the probability of future productivity jumps.

1.3 Why this changes how we read “misallocation”

In the misallocation literature, dispersion in MPK is often treated as evidence of inefficient allocation. This paper shows a mechanism where a substantial part of that dispersion may be *endogenously efficient* because it is linked to growth options. Reducing MPK dispersion by forcing firms to stop “investing in misallocation” can reduce aggregate productivity in the model.

2 Empirical environment: data, objects, portfolio construction, and jump definition

2.1 Data

The empirical work uses Compustat firm-year data (the main descriptive statistics and moment targets are computed over 1975–2021; some regressions use slightly shorter windows depending on leads/lags).

2.2 Key objects

The paper relies on three central firm-level objects:

- (a) **Total investment rate I/K** : includes both physical and intangible investment components (following Peters–Taylor style total capital logic).
- (b) **Marginal product of capital MPK**: an empirical proxy for the marginal revenue product of capital (exact construction details are in the paper’s Internet Appendix).

- (c) **Productivity** TFP: using firm-level TFP estimates from Imrohoroglu & Tuzel (2014), reported in industry-adjusted form.

Industry adjustment used repeatedly:

$$x_{i,t}^{(\text{ind. adj.})} = x_{i,t} - \text{median}\{x_{j,t} : j \in \text{same 2-digit SIC as } i\}.$$

2.3 Portfolio construction: the four $(I/K, \text{MPK})$ portfolios

Each year:

- (1) Within each 2-digit SIC industry, sort firms into *below-median* and *above-median* groups for I/K .
- (2) Independently sort into *below-median* and *above-median* groups for $\log \text{MPK}$.
- (3) This creates four portfolios:

$$(I/K_1, \text{MPK}_1), \quad (I/K_1, \text{MPK}_2), \quad (I/K_2, \text{MPK}_1), \quad (I/K_2, \text{MPK}_2),$$

where $(I/K_2, \text{MPK}_1)$ is the focal high-investment–low-MPK portfolio.

2.4 Young vs. mature firms

“Young” firms are those with age ≤ 10 years in Compustat (time since inclusion), and “mature” firms are the complement.

2.5 Definition of a jump (event classification)

A **jump** is a large joint increase in:

- sales (baseline: sales at least doubles), and
- MPK (baseline: at least a 50% increase).

To reduce noise, jumps are measured over a four-year window: growth from years $(t-1, t)$ to years $(t+1, t+2)$ is recorded as an event occurring in year t .

The paper also checks robustness to looser/tighter thresholds.

3 Reduced-form empirical models in the paper

This section collects the reduced-form models (regressions) used in the paper. These are distinct from the later structural model.

3.1 Model (9): linear probability model for jumps (Table 1)

The core reduced-form regression is a linear probability model (LPM) linking future jumps to current investment, current MPK, and age, with industry-year fixed effects.

A compact representation is:

$$\text{Jump}_{i,t+1} = \beta_0 + \beta_1 I/K_{i,t} + \beta_2 \log \text{MPK}_{i,t} + \beta_3 \log(\text{Age}_{i,t}) + \gamma_{\text{ind} \times t} + \varepsilon_{i,t}. \quad (1)$$

Variants include splitting I/K into physical vs. intangible components.

Table 1

Determinants of firm jumps.

	(1)	(2)	(3)	(4)	(5)
Physical I/K		0.020*** (11.20)	0.020*** (12.29)		
Intangible I/K		0.023*** (7.13)	0.031*** (9.97)		
Total I/K				0.050*** (15.24)	0.039*** (11.42)
Log MPK	-0.025*** (-18.49)		-0.027*** (-21.44)	-0.027*** (-21.15)	-0.027*** (-21.55)
Log age					-0.011*** (-18.09)
Ind $\times$ Year FE	x	x	x	x	x
R^2	0.043	0.031	0.055	0.053	0.057
N	203,253	203,054	203,054	203,253	203,253

Figure 1: Table1

Table 11

Effect of investment on firm and competitor growth.

	(1) $\log \frac{Sales_{i,t+5}}{Sales_{i,t}}$	(2) $\log \frac{K_{i,t+5}}{K_{i,t}}$	(3) $\log \frac{Prof_{i,t+5}}{Prof_{i,t}}$	(4) $\log TFP_{i,t+5}$
I/K_{firm}	0.418*** (10.36)	0.639*** (13.98)	0.373*** (7.11)	0.060*** (4.06)
$I/K_{comp \in (I/K_1, MPK_1)}$	-0.265 (-0.30)	1.333 (1.44)	0.083 (0.08)	0.083 (0.46)
$I/K_{comp \in (I/K_2, MPK_2)}$	0.571* (2.01)	1.412*** (5.70)	0.230 (0.72)	0.373*** (3.57)
$I/K_{comp \in (I/K_2, MPK_2)}$	-0.339** (-2.48)	-0.414*** (-3.40)	-0.301** (-2.19)	-0.054*** (-3.99)
$I/K_{comp \in (I/K_2, MPK_2)}$	0.416* (1.70)	0.437 (1.63)	0.335 (1.19)	0.086 (1.24)
Firm FE	x	x	x	x
R^2	0.015	0.088	0.011	0.004
N	134,860	134,860	126,492	90,533

Figure 2: Table11

3.2 Model (10): firm and competitor growth regressions (Table 11)

To test whether competitor investment matters for a firm's growth, the paper uses:

$$\Delta \log X_{i,t+5} = \beta I/K_{i,t} + \sum_{p \in \{(I/K_1, MPK_1), (I/K_1, MPK_2), (I/K_2, MPK_1), (I/K_2, MPK_2)\}} \gamma_p I/K_{p,t}^{\text{comp}} + \eta_{\text{ind} \times t} + u_{i,t}, \quad (2)$$

where X is (separately) Sales, Capital, Profitability, or TFP, and $I/K_{p,t}^{\text{comp}}$ is the *average investment rate of competitors in the same industry-year that are in portfolio p* . Industry-year fixed effects absorb common shocks.

3.3 Model (11): aggregate productivity growth predictive regressions (Table 12)

To link the investment pattern to aggregate productivity growth, the paper runs:

$$\log \left(\frac{TFP_{t+5}}{TFP_t} \right) = \alpha_0 + \sum_p \alpha_p \widetilde{I/K}_{p,t} + e_t, \quad (3)$$

where $\widetilde{I/K}_{p,t}$ is the *median* investment rate of firms in portfolio p at time t . This is done both as a bivariate regression (only the focal portfolio) and as a multivariate regression (all portfolios).

Table 12
Effect of investment on aggregate productivity.

	(1)	(2)	(3)	(4)	(5)
Panel A: Business sector TFP					
$I/K_{(I/K_1, MPK_1)}$	-0.013 (-0.07)				
$I/K_{(I/K_1, MPK_2)}$		0.091 (0.58)			
$I/K_{(I/K_2, MPK_1)}$			<u>0.091***</u> (8.95)		0.128*** (4.28)
$I/K_{(I/K_2, MPK_2)}$				<u>0.092***</u> (5.08)	-0.047 (-0.96)
Panel B: Utilization-adjusted TFP					
$I/K_{(I/K_1, MPK_1)}$	0.151 (0.63)				
$I/K_{(I/K_1, MPK_2)}$		0.263 (1.18)			
$I/K_{(I/K_2, MPK_1)}$			<u>0.116***</u> (7.21)		0.141*** (3.36)
$I/K_{(I/K_2, MPK_2)}$				<u>0.124***</u> (5.26)	-0.032 (-0.54)

Figure 3: Table12

Table 1
Determinants of firm jumps.

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Intangible I/K		0.023*** (7.13)	0.031*** (9.97)		
Total I/K				0.050*** (15.24)	0.039*** (11.42)
Log MPK	<u>-0.025***</u> (-18.49)		<u>-0.027***</u> (-21.44)	-0.027*** (-21.15)	-0.027*** (-21.55)
Log age					-0.011*** (-18.09)
Ind × Year FE	x	x	x	x	x
R^2	0.043	0.031	0.055	0.053	0.057
N	203,253	203,054	203,054	203,253	203,253

Figure 4: Table 1: Determinants of firm jumps.

4 Interpreting every table and figure (in logical order)

This section walks through each table/figure and states (i) what it measures and (ii) how it supports the paper's mechanism.

4.1 Table 1: determinants of firm jumps (LPM evidence)

What Table 1 does. It estimates the jump LPM (1) (with variations).

Table 1

Main empirical messages.

- **Investment predicts jumps:** higher I/K today is associated with higher probability of a jump next period. This holds for physical investment and intangible investment, and also for their sum (total I/K).
- **Conditional on investment, lower MPK predicts more jumps:** the coefficient on log MPK is negative and strongly significant, meaning that among firms with similar investment, the ones with lower current MPK are *more likely* to jump.
- **Younger firms jump more:** the coefficient on log(Age) is negative and significant.

Table 2

Descriptive statistics for total I/K and MPK sorted portfolios.

	$(I/K_1, MPK_1)$	$(I/K_1, MPK_2)$	$(I/K_2, MPK_1)$	$(I/K_2, MPK_2)$	$(I/K_2, MPK_1) - A/I$	
					Difference	t-stat
Panel A: Portfolio properties						
N	1428.3	993.8	997.2	1393.9		
Total I/K (median, ind. adj.)	-0.058	-0.048	0.085	0.097		
Log MPK (median, ind. adj.)	-0.40	0.35	-0.36	0.44		
Portfolio share	0.30	0.21	0.21	0.29		
Portfolio share among young firms (≤ 10 years)	0.22	0.14	0.27	0.37		
Age (median)	15.1	16.2	9.20	9.42	-2.96***	-4.65
Jump probability (%)	1.46	0.56	4.09	0.93	2.46***	8.25
Panel B: Innovative activity and product development						
Patents/K (mean)	9.24	6.67	26.5	17.0	11.9***	5.29
Patent value/K (mean)	24.9	22.8	90.1	93.0	32.4**	2.58
Patent Citations/K (mean)	293.9	142.1	1157.3	594.5	627.9***	4.04
Top 10% patents/K - 5 yr (mean)	1.17	0.47	4.40	2.20	2.40***	3.92
Top 10% patents/K - 10 yr (mean)	1.31	0.48	4.69	2.26	2.58***	3.62
Exposure to Life1 stage (median)	0.22	0.19	0.29	0.25	0.061***	13.2
Panel C: Productivity, returns, financial constraints						
Log TFP (median, ind. adj.)	-0.056	-0.033	0.020	0.074	0.020***	4.09
Log TFP (90th ptile, ind. adj.)	0.36	0.34	0.50	0.59	0.048**	2.33
Log future TFP (5yr later, median, ind. adj.)	-0.021	-0.019	0.012	0.010	0.015***	3.09
Log future TFP (5yr later, 90th ptile, ind. adj.)	0.41	0.35	0.55	0.50	0.10***	4.53
Excess future stock returns (VW mean, annual, %)	8.46	8.88	9.67	9.85	0.77	0.19
Total q (median)	0.43	0.44	0.86	0.87	0.27***	4.97
SA index (median)	-0.12	-0.26	0.21	0.14	0.21***	11.8
LW equity index (median)	0.026	-0.16	0.27	-0.027	0.27***	15.2

Figure 5: Table 2: descriptive statistics across $(I/K, MPK)$ portfolios

Interpretation for the paper’s narrative. The fact that *low current* MPK is associated with *higher jump likelihood* is inconsistent with a purely static view of MPK as a permanent firm attribute. It supports the idea that some firms temporarily look “over-capitalized” (low MPK) because they are building capacity for possible rapid growth.

4.2 Table 2: descriptive statistics across $(I/K, MPK)$ portfolios

What Table 2 does. It compares portfolio-level characteristics across the four portfolios and highlights what is special about $(I/K_2, MPK_1)$.

Panel A: portfolio properties.

- $(I/K_2, MPK_1)$ is a large share of firms (about one-fifth overall) and an even larger share among young firms.
- $(I/K_2, MPK_1)$ has by far the highest jump probability among the four portfolios.
- $(I/K_2, MPK_1)$ firms are younger than the overall sample.

Panel B: innovative activity and product development.

- $(I/K_2, MPK_1)$ shows substantially higher patenting intensity and citation-based measures.
- Higher exposure to the “Life1” stage (product development stage) is consistent with risky growth investments: firms must build capacity before fully learning product success, aligning with “investment predicts future jumps.”

Panel C: productivity, returns, and financial constraints.

- Productivity: high- I/K portfolios have higher current productivity, and within high- I/K , the high-MPK portfolio has higher current TFP. However, $(I/K_2, MPK_1)$ has *higher future TFP* (5 years later), especially in the right tail (e.g., 90th percentile), consistent with future “jumps”.

- Returns: realized excess returns are similar across portfolios, suggesting that the observed dispersion is not mainly driven by discount-rate differences across portfolios.
- Financial constraints: the table reports standard indices (SA, LW equity) to evaluate whether financial frictions alone explain the off-diagonal portfolios. The evidence does not support a simple “financial constraints only” story for why we see high investment with low MPK.

Interpretation. Table 2 builds the core empirical profile: $(I/K_2, MPK_1)$ is (i) young, (ii) innovation intensive, (iii) right-tail-growth prone, and (iv) not obviously explained by risk premia differences.

4.3 Figure 1: where the difference sits in the growth distribution

Figure 1

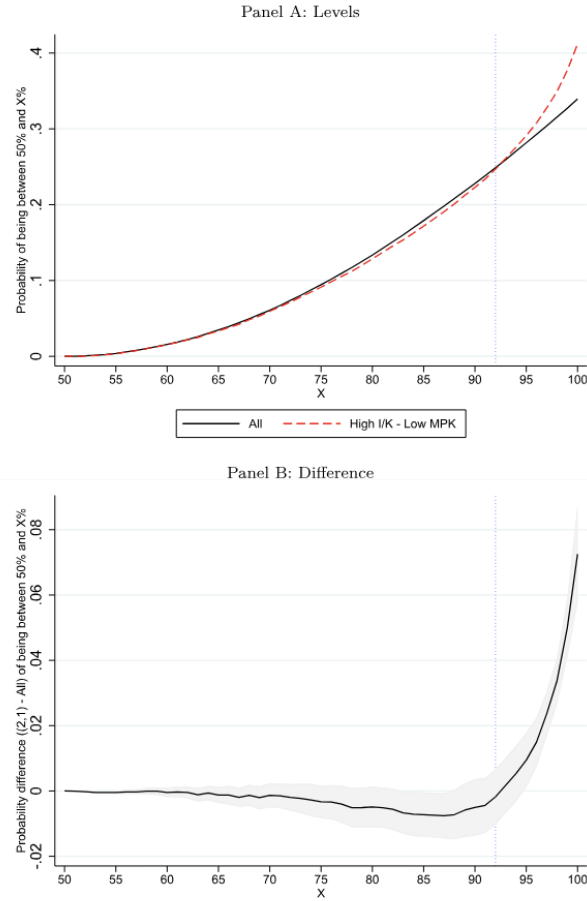


Figure 6: Figure 1: Distribution of above-median growth across firms.

What Figure 1 does. It compares the joint distribution of 2-year sales growth and 2-year MPK growth for all firms versus $(I/K_2, MPK_1)$ firms, focusing on the fraction of firms above progressively higher percentiles.

Main message. The two groups look similar in “normal” growth ranges (roughly up to the 90th percentile). The big difference is in the *upper tail*: $(I/K_2, MPK_1)$ firms are disproportionately represented among extreme growth events, which correspond to the jump cutoffs.

Interpretation. This is a key distributional fact supporting the model design: the paper needs a mechanism that specifically amplifies *rare, right-tail outcomes*, not just average growth.

4.4 Figure 2: time-series behavior of the $(I/K_2, MPK_1)$ portfolio

Figure 2

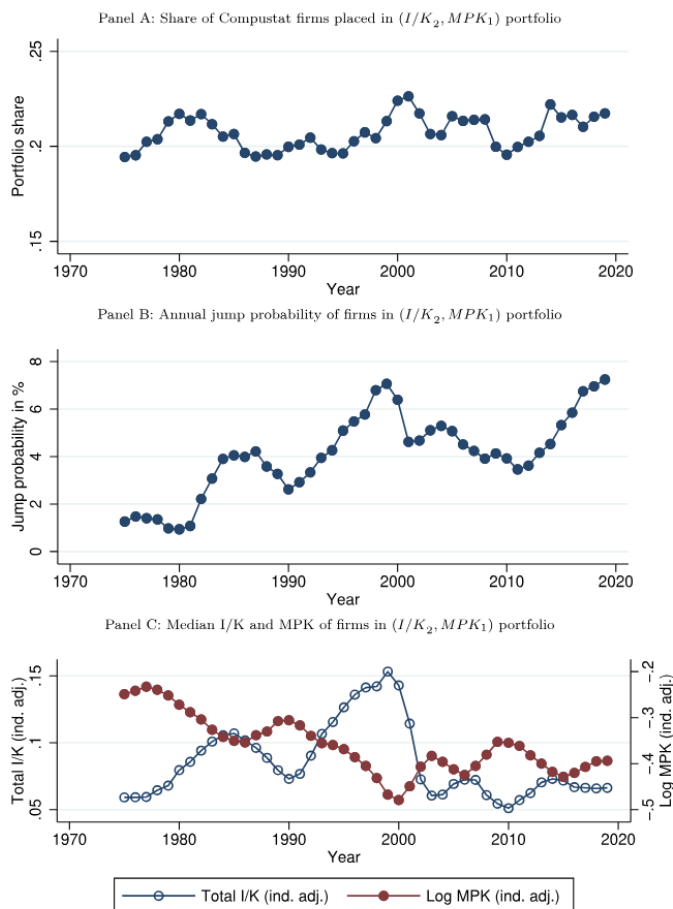


Figure 7: Figure 2: $(I/K_2, MPK_1)$ portfolio share and jump probability.

What Figure 2 does. It plots:

- (A) the share of firms in $(I/K_2, MPK_1)$ over time,
- (B) the annual jump probability within that portfolio,
- (C) the median industry-adjusted I/K and log MPK for that portfolio.

Main message. The portfolio share is fairly stable, while the jump probability varies over time and co-moves with the portfolio’s investment intensity and low-MPK behavior.

Interpretation. The time variation motivates a model where: (i) rare events matter, (ii) firms differ in exposure to those rare events, and (iii) investment is linked to that exposure.

4.5 Table 3: event study around jumps (mechanism timing)

What Table 3 does. It tracks median industry-adjusted I/K , median log MPK, and innovation measures from 5 years before a jump to 10 years after.

Table 3

Table 3
Jump event study.

	-5	-4	-3	-2	-1	0	1	2	3	4	6	8	10
Total I/K (median, ind. adj.)	0.029	0.035	0.035	0.032	0.037	0.073	0.076	0.077	0.064	0.044	0.015	0.006	-0.003
Log MPK (median, ind. adj.)	-0.514	-0.602	-0.638	-0.712	-0.931	-0.666	-0.112	0.105	0.035	-0.013	-0.058	-0.054	-0.097
Exposure to Life1 stage (median)	0.364	0.360	0.371	0.355	0.354	0.349	0.344	0.338	0.327	0.325	0.317	0.311	0.293
Patents/K (mean)	48.0	39.4	36.5	39.6	41.6	34.9	31.1	22.5	21.4	20.6	16.6	14.3	12.8
Patent value/K (mean)	164.0	166.7	148.8	176.7	177.0	233.4	227.6	184.3	160.9	165.1	184.4	161.2	167.7
Patent Citations/K (mean)	2735.9	2213.0	2123.3	2162.5	2829.5	2518.5	1795.7	1248.1	1132.6	957.7	684.3	453.1	388.7
Top 10% patents/K - 5 yr (mean)	6.7	8.0	8.4	8.4	10.8	10.2	7.4	5.5	4.7	4.8	3.1	1.8	1.7
Top 10% patents/K - 10 yr (mean)	8.3	9.5	8.5	10.2	13.1	11.5	8.2	6.7	5.1	4.9	3.0	2.5	2.1
Observations	1266	1420	1651	1949	2273	3234	3146	3070	2729	2405	1947	1574	1279

Figure 8: Table 3: Firm characteristics around jumps.

Timing patterns.

- **Investment ramps up pre-jump:** median $I/K^{(\text{ind. adj.})}$ rises notably in the years leading to the jump.
- **MPK is low pre-jump and spikes around jump:** log MPK^(ind. adj.) is depressed before the jump and then increases sharply around the event window.
- **Innovation proxies rise:** patent intensity and citation-based measures tend to be elevated around the jump window, consistent with product development/innovation being the underlying driver.

Interpretation. Table 3 is the most direct evidence for the “option value” story: firms look temporarily “misallocated” (high K relative to current sales / low MPK) while investing before a possible jump, and then the jump arrives, raising both sales and MPK.

5 Structural model: all equations, derivations, and key implications

5.1 Why a structural model is needed

The reduced-form evidence supports the idea that high- I/K , low-MPK firms are preparing for rare growth events. The structural model is designed to:

- generate a large $(I/K_2, \text{MPK}_1)$ portfolio endogenously,
- match the joint distribution of investment, growth, and jump events,
- quantify aggregate implications: what happens if firms stop “investing in misallocation”?

5.2 Technology and capital accumulation (Eqs. (1) and (3))

Firms have capital $K_{i,t}$ and productivity $Z_{i,t}$. Capital evolves as:

$$K_{i,t+1} = (1 - \delta)K_{i,t} + I_{i,t}. \quad (4)$$

Production is:

$$Y_{i,t} = Z_{i,t}^\alpha K_{i,t}^{1-\alpha}, \quad \alpha \in (0, 1). \quad (5)$$

5.3 Firm types and entry (Eq. (2))

Each entrant is born as either:

- **high-type** (can experience jumps), or
- **low-type** (no jumps; only Gaussian shocks).

Type at birth:

$$h_{i,t} \sim \text{Bernoulli}(p), \quad (6)$$

where p is the probability of being born high-type.

High-types transition to low-type at exogenous flow rate μ (absorbing low-type), and all firms exit with probability ϕ each period.

5.4 Productivity process with rare jumps (Eqs. (4)–(6))

Let $z_{i,t} = \log Z_{i,t}$.

$$z_{i,t+1} = z_{i,t} + \varepsilon_{i,t+1} + J_{i,t+1}, \quad \varepsilon_{i,t+1} \sim \mathcal{N}(0, \sigma^2). \quad (7)$$

Jump shock $J_{i,t+1}$:

$$\Pr(J_{i,t+1} = \zeta) = \lambda_{i,t}, \quad \Pr(J_{i,t+1} = 0) = 1 - \lambda_{i,t}, \quad (8)$$

where $\zeta > 0$ is the *jump size in log productivity*.

The jump probability depends on the firm's *scaled capital*:

$$k_{i,t} \equiv \frac{K_{i,t}}{Z_{i,t}}.$$

For **high-type firms**, jump probability is:

$$\lambda_{i,t} = \lambda_0 \left(\frac{k_{i,t}}{k_{ss}} \right)^\iota, \quad 0 < \iota < 1, \quad (9)$$

and for **low-type firms**, $\lambda_{i,t} = 0$ (no jumps).

Here k_{ss} is a normalization based on the no-jump steady state:

$$k_{ss} = \left(\frac{1 - \alpha}{R - 1 + \delta} \right)^{1/\alpha}. \quad (10)$$

5.5 Investment costs and the dynamic program (Eq. (7) and low-type analogue)

Per-period payoff equals output minus investment expenditure and adjustment costs. Adjustment costs are quadratic in the investment rate:

$$\text{AdjCost}(I, K) = \frac{c}{2} \left(\frac{I}{K} - \delta \right)^2 K.$$

Let $V_h(K, Z)$ be the value function for high-type firms. With discount factor $1/R$ and survival probability $(1 - \phi)$:

$$\begin{aligned} V_h(K_{i,t}, Z_{i,t}) = \max_{I_{i,t}} & \left\{ Z_{i,t}^\alpha K_{i,t}^{1-\alpha} - I_{i,t} - \frac{c}{2} \left(\frac{I_{i,t}}{K_{i,t}} - \delta \right)^2 K_{i,t} \right. \\ & \left. + \frac{1}{R}(1 - \phi) \left[(1 - \mu) \mathbb{E}_t[V_h(K_{i,t+1}, Z_{i,t+1})] + \mu \mathbb{E}_t[V_\ell(K_{i,t+1}, Z_{i,t+1})] \right] \right\}, \end{aligned} \quad (11)$$

subject to (4) and (7)–(9).

Low-type firms solve the same problem but with $\mu = 0$ (stay low-type) and $\lambda_{i,t} = 0$ (no jumps):

$$\begin{aligned} V_\ell(K_{i,t}, Z_{i,t}) = \max_{I_{i,t}} & \left\{ Z_{i,t}^\alpha K_{i,t}^{1-\alpha} - I_{i,t} - \frac{c}{2} \left(\frac{I_{i,t}}{K_{i,t}} - \delta \right)^2 K_{i,t} \right. \\ & \left. + \frac{1}{R}(1 - \phi) \mathbb{E}_t[V_\ell(K_{i,t+1}, Z_{i,t+1})] \right\}. \end{aligned} \quad (12)$$

5.6 Derivation 1: MPK in the model and why $k = K/Z$ is the key state

From (5):

$$\text{MPK}_{i,t} \equiv \frac{\partial Y_{i,t}}{\partial K_{i,t}} = (1 - \alpha) Z_{i,t}^\alpha K_{i,t}^{-\alpha}.$$

Now rewrite using $K_{i,t} = k_{i,t} Z_{i,t}$:

$$\text{MPK}_{i,t} = (1 - \alpha) Z_{i,t}^\alpha (k_{i,t} Z_{i,t})^{-\alpha} = (1 - \alpha) k_{i,t}^{-\alpha}.$$

Key implication: in this model, MPK depends *only* on scaled capital k , not on Z separately. Thus:

- higher $k \Rightarrow$ lower current MPK,
- higher k (for high-types) $\Rightarrow$ higher jump probability $\lambda(k)$ via (9).

This is the formal reason the model can generate firms that are simultaneously high-investment and low-MPK: they choose high k to raise $\lambda(k)$ even though that lowers current MPK.

5.7 Derivation 2: steady-state normalization k_{ss}

In a no-jump, no-growth benchmark, the steady-state optimality condition is the standard user-cost condition:

$$\text{MPK} = R - 1 + \delta.$$

Using $\text{MPK} = (1 - \alpha)k^{-\alpha}$:

$$(1 - \alpha)k_{ss}^{-\alpha} = R - 1 + \delta \quad \Rightarrow \quad k_{ss} = \left(\frac{1 - \alpha}{R - 1 + \delta} \right)^{1/\alpha},$$

which matches (10).

5.8 Derivation 3: marginal cost of investment under the adjustment cost

Let $i \equiv I/K$. The investment cost term is:

$$I + \frac{c}{2}(i - \delta)^2 K.$$

Differentiate w.r.t. I :

$$\frac{\partial}{\partial I} \left[I + \frac{c}{2}(i - \delta)^2 K \right] = 1 + c(i - \delta).$$

This implies that in the optimality condition, marginal benefit of raising K_{t+1} (including its effect on future output and future jump probabilities) is equated to $1 + c(i - \delta)$.

5.9 Mechanism summary (why the model matches the data)

- (1) High-type firms have an *endogenous* jump probability increasing in $k = K/Z$.
- (2) Therefore, high-type firms invest beyond what would be justified by current production alone, to raise k and thus raise $\lambda(k)$.
- (3) That extra investment lowers contemporaneous MPK (because $\text{MPK} = (1 - \alpha)k^{-\alpha}$), producing a high- I/K / low-MPK portfolio.
- (4) When jumps occur, productivity rises sharply (via ζ), which raises sales and MPK jointly, matching the empirical definition of jumps.

6 Estimation: targeted moments and parameter interpretation

6.1 Estimation method

The paper uses indirect inference / simulated method of moments logic: choose parameters θ so that simulated data matches empirical moments. A generic objective is:

$$\hat{\theta} \in \arg \min_{\theta} \left(m^{\text{data}} - m^{\text{sim}}(\theta) \right)^{\top} W \left(m^{\text{data}} - m^{\text{sim}}(\theta) \right),$$

with m containing the moments listed in Table 4.

6.2 Table 4: what moments are targeted and what they identify

Table 4 and Table 5

Table 4
Moment conditions.

	Data	Baseline
Panel A: General moments		
<u>IQR of I/K among young firms</u>	0.252	0.226
<u>IQR of I/K among mature firms</u>	0.092	0.118
<u>IQR of sales growth</u>	0.243	0.213
Median I/K	0.164	0.164
Median profitability	0.370	0.389
Panel B: Moments related to jump realizations		
<u>Jump probability of young firms</u>	0.028	0.023
<u>Median sales jump size</u>	2.966	2.595
Median log MPK jump size	0.720	0.665
Median jump age	6.000	6.000
Panel C: Moments for the $(I/K_2, MPK_1)$ portfolio		
Portfolio share	0.207	0.187
Jump probability	0.041	0.050
I/K (ind. adj.)	0.086	0.097
Portfolio share among young firms	0.269	0.309

This table presents the target empirical moments and model-generated moments obtained from the estimation of our baseline and counterfactual models. The empirical moments, presented in column 1, are computed using Compustat data from the period 1975–2021. For more details, please refer to Internet Appendix I.2. The parameter estimates for the benchmark model can be found in Table 5.

Table 5
Parameter estimates.

Panel A: Calibrated parameters	
Discount rate, $1/R$	0.91
Exit probability, φ	0.031
Panel B: Estimated parameters	
Production function curvature, α	0.361 (0.004)
Capital depreciation rate, δ	0.196 (0.001)
Gaussian shock volatility, σ	0.328 (0.003)
Adjustment cost parameter, c	2.914 (0.023)
Jump probability level, λ_0	0.031 (0.001)
Jump probability curvature, ι	0.484 (0.006)
<u>Jump size, ζ</u>	<u>2.187</u> (0.017)
<u>Type switching probability, μ</u>	<u>0.101</u> (0.002)
<u>Probability of being born high-type, p</u>	<u>0.928</u> (0.020)

This table provides the parameters resulting from the estimation of the baseline model. Panel A displays the calibrated parameters, while Panel B presents the estimated parameters, with the target and model-generated moments shown in Table 4. The standard errors for the parameters are presented in parentheses.

Figure 9: Tables 4–5: Moment conditions and baseline parameter estimates.

Panel A (general moments). These discipline dispersion and central tendency:

- IQR of I/K among young and mature firms (investment dispersion + life-cycle difference),
- IQR of sales growth (output dispersion),

- median I/K and median profitability (levels).

Identification intuition:

- σ increases dispersion in both output and investment;
- c dampens investment volatility more than sales volatility;
- α and δ are pinned down by profitability and median investment.

Panel B (jump realizations). These identify the jump mechanism:

- jump probability of young firms helps pin down (λ_0, ι, p) ,
- median jump sizes in sales and log MPK discipline jump size ζ (given α),
- median jump age helps discipline (λ_0, μ) (jumps stop once firms switch to low-type).

Panel C (moments for $(I/K_2, \text{MPK}_1)$). These discipline how large the high-investment–low-MPK group is, which is crucial for pinning down the jump probability function:

- portfolio share (overall and among young),
- jump probability within the portfolio,
- median industry-adjusted I/K within the portfolio.

6.3 Table 5: baseline parameter estimates (economic meaning)

Table 5 reports:

- calibrated: discount factor $1/R$ and exit probability ϕ ,
- estimated: $(\alpha, \delta, \sigma, c, \lambda_0, \iota, \zeta, \mu, p)$.

Interpretation highlights:

- (1) A large p implies most entering public firms are high-type (consistent with IPO selection).
- (2) μ governs how fast high-type advantage disappears; it affects expected duration of being eligible for jumps.
- (3) ζ is chosen so that the model generates sales and MPK increases consistent with the empirical jump definition.

7 Model validation and counterfactuals (all tables/figures in this part)

7.1 Why the “no-jump” benchmark fails (Table 6 and Table 7)

No-jump model. Set $\lambda = 0$ for everyone; then the model is a standard neoclassical investment model with random-walk productivity and adjustment costs.

Table 6 (moments) and Table 7 (parameters). The no-jump model can be tuned to match some dispersion moments, but it fails in the key dimension: the $(I/K_2, \text{MPK}_1)$ portfolio becomes essentially empty (near 0% vs. about 20% in data), and jump-related moments are either zero or can only be mimicked by extreme Gaussian shocks that distort other implications.

Table 6

Table 6**Moments in a model with no jumps.**

	Data	Targeting Panel A	Targeting Panel A & B
Panel A: General moments			
IQR of I/K among young firms	0.252	0.090	0.139
IQR of I/K among mature firms	0.092	0.090	0.140
IQR of sales growth	0.243	0.165	0.265
Median I/K	0.164	0.156	0.138
Median profitability	0.370	0.425	0.419
Panel B: Moments related to jump realizations			
Jump probability of young firms	0.028	0.000	0.000
Median sales jump size	2.966	n/a	2.057
Median log MPK jump size	0.720	n/a	0.410
Median jump age	6.000	n/a	11.000
Panel C: Moments for the $(I/K_2, MPK_1)$ portfolio			
Portfolio share	0.207	0.001	0.002
Jump probability	0.041	0.000	0.000
I/K (ind. adj.)	0.086	0.001	0.001
Portfolio share among young firms	0.269	0.001	0.002

Figure 10: Table 6: Portfolio dynamics around jumps and no-jump moments.

Table 7

Table 7**Parameter estimates in a model with no jumps.**

	Targeting Panel A	Targeting Panel A & B
Production function curvature, α	0.310 (0.003)	0.280 (0.002)
Capital depreciation rate, δ	0.224 (0.002)	0.318 (0.002)
Gaussian shock volatility, σ	0.359 (0.004)	0.611 (0.005)
Adjustment cost parameter, c	4.605 (0.100)	3.202 (0.032)

Figure 11: Table 7: Parameter estimates in a model with no jumps.

Interpretation. Without jumps linked to future upside, the model generates a counterfactually tight link between I/K and MPK: high I/K tends to coincide with high MPK, eliminating the off-diagonal portfolio that the paper tries to explain.

7.2 Counterfactual: firms ignore the possibility of jumps (Table 8, Table 9, Fig. 4, Table 10, Fig. 5)

The key counterfactual keeps the baseline environment/parameters but changes behavior: firms choose investment as if jumps will not occur (ignoring the upside channel that motivates “investing in misallocation”).

Table 8: baseline vs counterfactual moments Main differences:

- Investment dispersion (especially among young firms) collapses.
- The $(I/K_2, MPK_1)$ portfolio share shrinks dramatically, particularly among young firms.

- Jump-related behavior becomes inconsistent with the empirical targeted patterns.

Table 9: predicting jumps in the model This table runs the same predictive regression logic as Table 1 on simulated data.

- Baseline model matches the *sign pattern* in the data: I/K positively predicts jumps, while $\log MPK$ negatively predicts jumps.
- Counterfactual flips the signs: if firms ignore jumps, high investment and low MPK no longer predict jump realizations in the simulated data.

Figure 4 and Tables 8–9

Table 8

Moment conditions: Baseline and Counterfactual.

	Data	Baseline	Counterfactual
Panel A: General moments			
IQR of I/K among young firms	0.252	0.226	0.136
IQR of I/K among mature firms	0.092	0.118	0.103
IQR of sales growth	0.243	0.213	0.186
Median I/K	0.164	0.164	0.160
Median profitability	0.370	0.389	0.488
Panel B: Moments related to jump realizations			
Jump probability of young firms	0.028	0.023	0.011
Median sales jump size	2.966	2.595	2.435
Median log MPK jump size	0.720	0.665	0.747
Median jump age	6.000	6.000	5.000
Panel C: Moments for the $(I/K_2, MPK_1)$ portfolio			
Portfolio share	0.207	0.187	0.000
Jump probability	0.041	0.050	n/a
I/K (ind. adj.)	0.086	0.097	n/a
Portfolio share among young firms	0.269	0.309	0.000

This table presents the target empirical moments and model-generated moments obtained from the estimation of our baseline and counterfactual models. The empirical moments, presented in column 1, are computed using Compustat data from the period 1975–2021. For more details, please refer to Internet Appendix I.2. The parameter estimates for the benchmark model can be found in Table 5. The counterfactual model, as discussed in Section 4.2.2, adopts the same parameter estimates as the baseline model, except that firms’ investment policy ignores the possibility of jumps.

Table 9

Predicting jumps in the model.

	Data	Baseline	Counterfactual	No Jump Targeting A	No Jump Targeting A & B
Total I/K	0.050	0.112 [0.085; 0.141]	−0.034 [−0.084; 0.029]	0.000 [0.000; 0.000]	0.002 [0.000; 0.021]
Log MPK	−0.027	−0.062 [−0.079; −0.046]	0.015 [−0.010; 0.036]	0.000 [0.000; 0.000]	−0.000 [−0.004; 0.000]

This table presents coefficient estimates obtained from linear probability models analyzing realized jumps using both empirical data and model-simulated data. The dependent variable is the jump dummy, which takes a value of 1 when a jump occurs from the current period to the next period. Jumps are defined as cases where a company’s sales double, accompanied by a minimum 50% increase in its MPK. Column 1 reproduces Column 4 of Table 1, which contains coefficient estimates based on empirical data. Columns 2 to 5 correspond to coefficient estimates obtained from simulations of the Baseline model, the counterfactual model, and two versions of the no-jump model, respectively. Point estimates for the coefficients are simulation averages, while the confidence intervals (presented in parentheses) are constructed from the 2.5th and 97.5th percentiles of the simulated distribution of each coefficient.

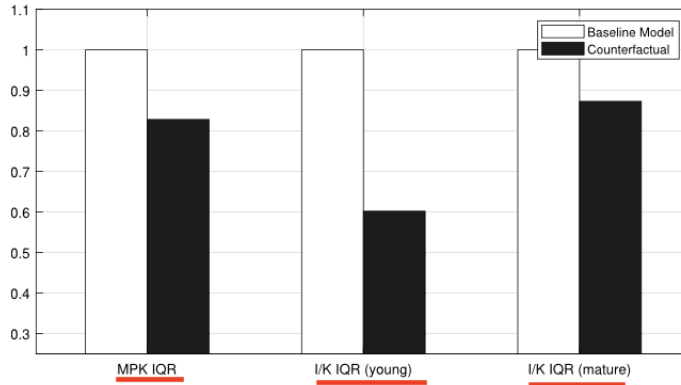


Fig. 4. Dispersion in the Model.

The figure displays the interquartile range (IQR) for MPK and the investment rate IQR separately for young and mature firms. In order to facilitate presentation and comparisons, the values in the baseline model are normalized to 1.

Figure 12: Figure 4 and Tables 8–9: Counterfactual moments and predicting jumps in the model.

Figure 4: dispersion changes Figure 4 summarizes the dispersion (IQR) changes relative to baseline:

- MPK dispersion declines (mechanically: firms stop holding high k states),
- investment dispersion among young firms drops sharply (because the “option value” motive disappears).

Table 10: model-generated MPK dispersion across portfolios This table compares the within-portfolio MPK gaps (e.g., high-MPK minus low-MPK groups) across data and models. The counterfactual often yields missing/off-diagonal categories because the $(I/K_2, MPK_1)$ group becomes tiny.

Figure 5 and Table 10

Table 10

Model-generated MPK dispersion across portfolios.

	Data	Baseline	Counterfactual	No Jump Targeting A	No Jump Targeting A & B
$(I/K_1, MPK_1)$	-0.40	-0.22	-0.18	-0.20	-0.29
$(I/K_1, MPK_2)$	0.35	0.13	n/a	0.02	-0.00
Difference	0.75	0.35	n/a	0.22	0.29
$(I/K_2, MPK_1)$	-0.36	-0.24	n/a	0.01	-0.01
$(I/K_2, MPK_2)$	0.44	0.34	0.19	0.19	0.27
Difference	0.80	0.58	n/a	0.19	0.28

This table compares the median MPK for I/K and MPK Sorted Portfolios using both empirical data and model-simulated data. Column 1 displays the median industry-adjusted MPK values for the portfolios, along with the MPK dispersion for the low and high investment groups. Columns 2 to 5 present the same median-adjusted statistics obtained from simulations of different models. Column 2 represents the Baseline model, Column 3 represents the counterfactual model, and Columns 4 and 5 represent two versions of the no-jump model.

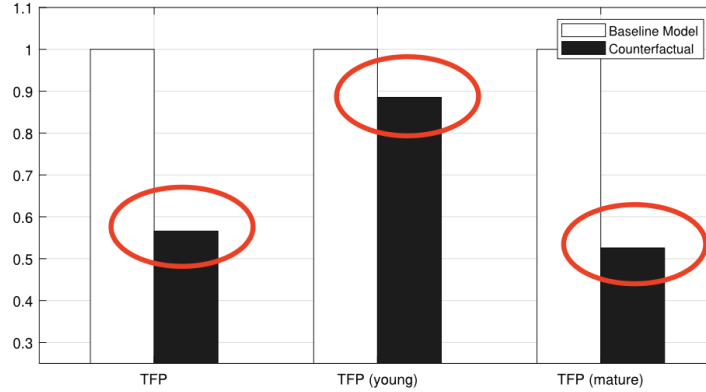


Fig. 5. Aggregate Productivity in the Model.

The figure displays the aggregate productivity level for all firms as well as for young and mature firms separately. The logarithm of aggregate productivity level is computed as a Solow residual, obtained by subtracting $(1 - \alpha)$ times the log total capital from the log total output. In Panel B, the investment rate IQR is displayed separately for young and mature firms. In Panel C, productivity obtained from the Solow residual is displayed separately for young and mature firms obtained using the total output and capital for each group. In order to facilitate presentation and comparisons, the values in the baseline model are normalized to 1.

Figure 13: Figure 5 and Table 10: Aggregate productivity and MPK dispersion across portfolios.

Figure 5: aggregate productivity consequences Aggregate productivity is computed as a Solow residual in the model:

$$\log TFP_t^{\text{Solow}} = \log Y_t - (1 - \alpha) \log K_t.$$

Figure 5 shows a large decline in aggregate productivity when firms stop investing in misallocation. Interpretation: eliminating the mechanism reduces MPK dispersion but also reduces the frequency/impact of productivity jumps, lowering aggregate productivity.

7.3 Additional empirical support: competitor effects (Table 11)

Table 11 uses regression (2).

Table 11

Table 11
Effect of investment on firm and competitor growth.

	(1) $\log \frac{Sales_{i,t+5}}{Sales_{i,t}}$	(2) $\log \frac{K_{i,t+5}}{K_{i,t}}$	(3) $\log \frac{Profit_{i,t+5}}{Profit_{i,t}}$	(4) $\log TFP_{i,t+5}$
I/K_{firm}	0.418*** (10.36)	0.639*** (13.98)	0.373*** (7.11)	0.060*** (4.06)
$I/K_{comp \in (I/K_1, MPK_1)}$	-0.265 (-0.30)	1.333 (1.44)	0.083 (0.08)	0.083 (0.46)
$I/K_{comp \in (I/K_1, MPK_2)}$	0.571* (2.01)	1.412*** (5.70)	0.230 (0.72)	0.373*** (3.57)
$I/K_{comp \in (I/K_2, MPK_1)}$	-0.339** (-2.48)	-0.414*** (-3.40)	-0.301** (-2.19)	-0.054*** (-3.99)
$I/K_{comp \in (I/K_2, MPK_2)}$	0.416* (1.70)	0.437 (1.63)	0.335 (1.19)	0.086 (1.24)
Firm FE	x	x	x	x
R^2	0.015	0.088	0.011	0.004
N	134,860	134,860	126,492	90,533

Figure 14: Table 11: Effect of investment on firm and competitor growth.

Key reading:

- A firm's own I/K is positively related to its future growth (sales, capital, profitability, and TFP).
- Competitor investment in the *high- I/K , low-MPK* portfolio tends to have negative coefficients for a firm's subsequent growth, consistent with business-stealing / competitive pressure effects.

This is complementary evidence: growth-oriented investment may create competitive externalities, but it can still raise aggregate productivity if it reallocates capacity toward high-type firms that are more likely to achieve major productivity improvements.

7.4 Aggregate evidence: investment in $(I/K_2, MPK_1)$ and aggregate TFP growth (Table 12)

Table 12 estimates (3). Main message: investment intensity in $(I/K_2, MPK_1)$ predicts higher future aggregate TFP growth, even controlling for investment in the other portfolios (multivariate specifications), and under alternative TFP constructions (Solow residual vs utilization-adjusted TFP).

8 Takeaway list

- T1:** High- I/K , low-MPK firms are common and disproportionately experience rare “jumps”.
- T2:** The jump probability is empirically predictable by investment and (negatively) by current MPK.
- T3:** A structural model with two types and endogenous jump probability $\lambda(k)$ explains why firms may rationally hold high capital relative to current productivity:

$$MPK = (1 - \alpha)k^{-\alpha} \downarrow \text{ in } k, \quad \lambda(k) = \lambda_0(k/k_{ss})^{\iota} \uparrow \text{ in } k.$$

- T4:** “Removing misallocation” by shutting down this mechanism reduces MPK dispersion but reduces aggregate productivity substantially in the model.

T5: Empirically, the investment in the high- I/K , low-MPK portfolio predicts higher aggregate productivity growth.